

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Simulation-Based Assessment Question

Duration: 1hr

- 1. Simulate a Partially Observable Markov Decision Process (POMDP) for autonomous vehicle navigation under limited visibility.**

Objective:

Simulate an autonomous vehicle navigating a road when visibility is limited due to fog, rain, or obstacles.

POMDP Components:

- **States (S):** Clear Road, Obstacle Ahead, Vehicle Ahead, Left Turn, Right Turn
- **Actions (A):** Move Forward, Turn Left, Turn Right, Brake
- **Observations (O):** Clear, Obstacle Detected, Vehicle Detected, Uncertain
- **Transition Model:** Represents the probability of moving from one state to another after an action.
- **Observation Model:** Represents the probability of receiving an observation given the actual state.
- **Reward:** Positive reward for safe movement and reaching the destination; negative reward for collisions, unsafe actions, or unnecessary braking.
- **Belief State:** Maintains the probability distribution over possible states because the vehicle cannot directly observe the environment.

Simulation Flow:

Environment → Partial Observation → Belief State Update → Action Selection → State Transition → Reward

Step 1 — Understand what you are simulating

Imagine a car moving on a road with fog.

The actual road condition can be:

- Clear
- Obstacle
- Vehicle

But because of limited visibility, the car **cannot directly know the actual state**. It only gets observations such as:

- Clear
- Obstacle Detected
- Uncertain

The car maintains a **belief** (probability) about what the current state might be and chooses an action based on that belief.

Step 2 — Install the required library

In Google Colab, run:

```
!pip install numpy matplotlib
```

Step 3 — Use this complete simulation

```
import numpy as np
import matplotlib.pyplot as plt

states = ["Clear", "Obstacle", "Vehicle"]
actions = ["Forward", "Brake", "Left", "Right"]
observations = ["Clear", "Obstacle Detected", "Uncertain"]

belief = np.array([0.7, 0.2, 0.1])

transition = {
    "Forward": np.array([
        [0.8, 0.15, 0.05],
        [0.1, 0.8, 0.1],
        [0.2, 0.1, 0.7]
    ]),
    "Brake": np.array([
        [0.9, 0.05, 0.05],
        [0.05, 0.9, 0.05],

```

```

    [0.1, 0.1, 0.8]
]),
"Left": np.array([
    [0.7, 0.2, 0.1],
    [0.2, 0.7, 0.1],
    [0.3, 0.1, 0.6]
]),
"Right": np.array([
    [0.7, 0.1, 0.2],
    [0.2, 0.7, 0.1],
    [0.3, 0.1, 0.6]
])
}
observation_prob = {
    "Clear": np.array([0.9, 0.05, 0.05]),
    "Obstacle Detected": np.array([0.05, 0.9, 0.1]),
    "Uncertain": np.array([0.05, 0.05, 0.85])
}
rewards = {
    "Forward": [10, -20, 5],
    "Brake": [-2, 8, 5],
    "Left": [5, -10, 3],
    "Right": [5, -10, 3]
}
history = []
for step in range(15):

    expected_rewards = {}

    for action in actions:
        expected_rewards[action] = np.dot(belief, rewards[action])

    action = max(expected_rewards, key=expected_rewards.get)

    predicted_belief = belief @ transition[action]

    observation = np.random.choice(
        observations,
        p=observation_prob["Clear"] * predicted_belief +

```

```

        observation_prob["Obstacle Detected"] * predicted_belief +
        observation_prob["Uncertain"] * predicted_belief
    )

    if observation == "Clear":
        likelihood = observation_prob["Clear"]
    elif observation == "Obstacle Detected":
        likelihood = observation_prob["Obstacle Detected"]
    else:
        likelihood = observation_prob["Uncertain"]

    belief = predicted_belief * likelihood
    belief = belief / belief.sum()

    history.append([
        step + 1,
        action,
        observation,
        belief[0],
        belief[1],
        belief[2]
    ])

print("Step | Action | Observation | Clear | Obstacle | Vehicle")print("-" * 65)
for row in history:
    print(
        f"{row[0]:4} | "
        f"{row[1]:7} | "
        f"{row[2]:18} | "
        f"{row[3]:.2f} | "
        f"{row[4]:.2f} | "
        f"{row[5]:.2f}"
    )

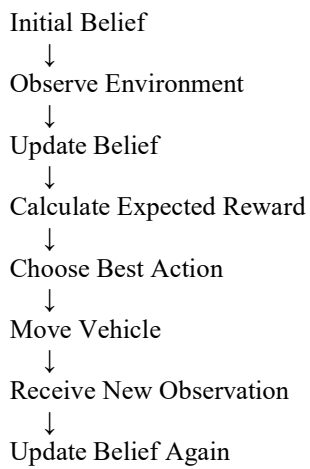
steps = [x[0] for x in history]clear = [x[3] for x in history]obstacle = [x[4] for x in history]vehicle = [x[5] for x in
history]

plt.plot(steps, clear, marker="o", label="Clear")plt.plot(steps, obstacle, marker="o",
label="Obstacle")plt.plot(steps, vehicle, marker="o", label="Vehicle")
plt.xlabel("Simulation Step")plt.ylabel("Belief Probability")plt.title("POMDP Autonomous Vehicle
Navigation")plt.legend()plt.grid()plt.show()

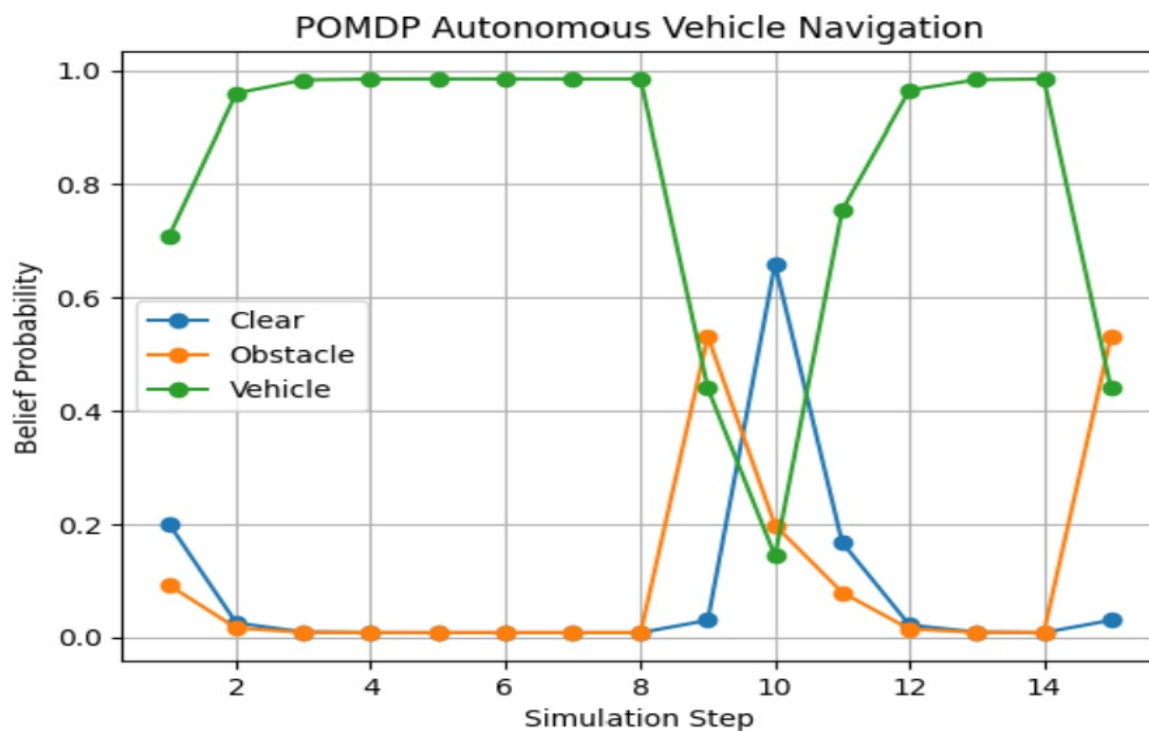
```

Step 4 — What happens in the program?

The important part is:



Step	Action	Observation	Clear	Obstacle	Vehicle
1	Forward	Uncertain	0.20	0.09	0.71
2	Brake	Uncertain	0.02	0.02	0.96
3	Brake	Uncertain	0.01	0.01	0.98
4	Brake	Uncertain	0.01	0.01	0.98
5	Brake	Uncertain	0.01	0.01	0.98
6	Brake	Uncertain	0.01	0.01	0.98
7	Brake	Uncertain	0.01	0.01	0.98
8	Brake	Uncertain	0.01	0.01	0.98
9	Brake	Obstacle Detected	0.03	0.53	0.44
10	Brake	Clear	0.66	0.20	0.14
11	Forward	Uncertain	0.17	0.08	0.75
12	Brake	Uncertain	0.02	0.01	0.96
13	Brake	Uncertain	0.01	0.01	0.98
14	Brake	Uncertain	0.01	0.01	0.98
15	Brake	Obstacle Detected	0.03	0.53	0.44



Code of Conduct:

I P. Sahith Sai (192425029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Sahith Sai

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively